ECE 198 Project: Scalability of Exercise as a Powerful Healthcare Tool for Geriatric Patients

Congratulations! You have recently been hired by the Centre for Community, Clinical & Applied Research Excellence (CCCARE) and have been tasked with streamlining some of their daily operations. CCCARE'S mission is to enhance the health of people in the community and support individuals as they manage illness or recovery from injury. They run general exercise programs, as well as specialized sessions for people undergoing or recovering from cancer treatment, stroke survivors, and patients who have completed earlier programs but wish to continue using the facility.

Your team's challenge is to develop ways for CCARE to improve how they deliver care and scale their services. You've been selected for your strong problem-solving skills and your ability to innovate within an interdisciplinary team. CCCARE has identified three focus areas for potential improvements:

- **Expanding and managing the individualized exercise plans for each participant**
- **Support remote, at-home care to reach more people and help graduates of the programs transition to self-managed exercise**
- **Tracking and assessing exercise activity to ensure quality and progress for all participants**

You are encouraged to explore **one or more of these focus areas** to develop your own innovative solution related to exercise for older adults.

Sub Problem 1

Problem Space: Scale Personalized Care
At CCCARE, participants receive a personalized exercise plan (referred to as a "dosage") which specifies the amount and type of physical activity designed to meet their needs – this plan is constantly adjusted. Initial assessments carried out by Kinesiologists, rehabilitation, or other health professionals determine the starting point for each participant based on their current physical ability, medical history, and personal goals.

Patients' exercise history and data must be consulted when adjusting exercise plans. Currently, CCCARE stores data using a mix of digital and paper-based systems: exercise logs are stored on paper, appointments are scheduled using AirTable, intake assessments are written on paper and manually input into a Microsoft Access database by staff, and referrals from Waterloo Regional Health Network (WRHN) are sent in by fax which requires manual entry into the database. Unfortunately, Access cannot be

integrated with other clinical software or visualize data, hindering data sharing and coordination.

More information on dosages & exercise sessions can be found in the “Dosages and Exercise Sessions” file on Learn.

The Challenge: Design a **scalable** system that makes it easier to prescribe, adjust, store, and share exercise doses efficiently.
Resources:

- Github tutorial on how to create a database

Sub Problem 2

Problem Space: Exercise Tracking and Measurement
In CCCARE, an instructor will work with 10-12 patients in a session. Meanwhile, graduated participants can have up to 30 people in a session with one instructor. Due to minimal supervision, there is no easy way to track each patient as they exercise, therefore some participants may not complete their full dose, while others might overdo it. Student volunteers are available to assist participants who may require extra support, such as patients with dementia and stroke patients. However, as not all participants will have a volunteer working with them one-on-one, there is a need to track the quality of exercises performed so that instructors are able to evaluate the participants’ progress.

More information on dosages & exercise sessions can be found in the “Dosages and Exercise Sessions” file on Learn.

The Challenge: Design a system that **tracks** and **measures** the quality of patient activity during the exercise.

Sub Problem 3

Problem Space: Remote (At Home) Care
CCCARE wants to provide remote care for patients because the in-person programs have limited capacity, and they would like to encourage graduated patients to rely less on the facility. Some patients on the waitlist cannot physically attend due to medical constraints, leading to less than 10% of stroke and cancer patients from GRH receiving treatment at CCCARE. Moving towards remote treatment allows for the treatment of far more people at once without worrying about space or class size and provides graduates with support and community as they shift to more independent exercise. However,

CCCARE is concerned with ensuring participants are getting safe and effective exercise.

The Challenge: Design a system that ensures patients are completing the correct exercises and adhering to their treatment plans **at home**.

Here are some things to think about as you come up with solutions for this problem space: How can we combat that feeling of isolation and support them in maintaining their exercise habits? They've been used to working out with a group and an instructor, so how do we keep that sense of community going after they're on their own? How can we track their exercise and whether they are doing the right ones and in correct dosage without breaching privacy?

Deliverables

Project Pitch Video Presentation – Due Friday, Oct. 19

- 5-minute, pre-recorded presentation uploaded to Learn
    - This can be a video, or you can include speech annotations on a presentation
    - MS PowerPoint can do this natively. Instructions are available online: https://support.microsoft.com/en-us/office/record-a-slide-show-with-narration-and-slide-timings-0b9502c6-5f6c-40ae-b1e7-e47d8741161c
- Needs to describe:
    - What part of the problem is the team seeking to address?
    - A quick overview of 2 concept solutions
    - The hardware required to complete the project

In-person Project Update Presentation –week of Nov. 9 (dates/times TBA)

- 5-minute presentation on project progress + 5-minute Q&A and feedback with the teaching team
- Bring your prototype, and you can present on a laptop/tablet
- The teaching team will be looking for progress, but we aren't expecting a finished project.
- This is also a chance to ask questions and seek advice!

Final Project Symposium – Monday, November 23 (8-Stream) and Wednesday, November 25 (4-Stream)

- Demonstrate your project prototype to stakeholders, the teaching team, and each other!
- The symposium will run in the Ideas Clinic (PSE-1427)
    - More details will be available as we get closer